

3. Indicator species may be used in an investigation to monitor water pollution at three different locations along a stream.

- (a) The method for an investigation to monitor water pollution in a stream is described below. However, some of the stages are missing.

Complete the method by writing in the missing steps.

[2]

1. Take a tray and a net to the first location.
2.
3. Study the organisms in the tray.
4.
5. Pour the invertebrates and the water gently back into the stream.
6. Repeat steps 1–5 at the second and third locations.



(b) The following chart is available to you.

Pollution level

Indicator species

A Clean water



Stonefly nymph



Mayfly nymph

B Some pollution



Freshwater shrimp



Caddis fly larva

C Moderate pollution



Water louse



Bloodworm

D High pollution



Sludgeworm



Rat-tailed maggot

E Very High pollution – no life

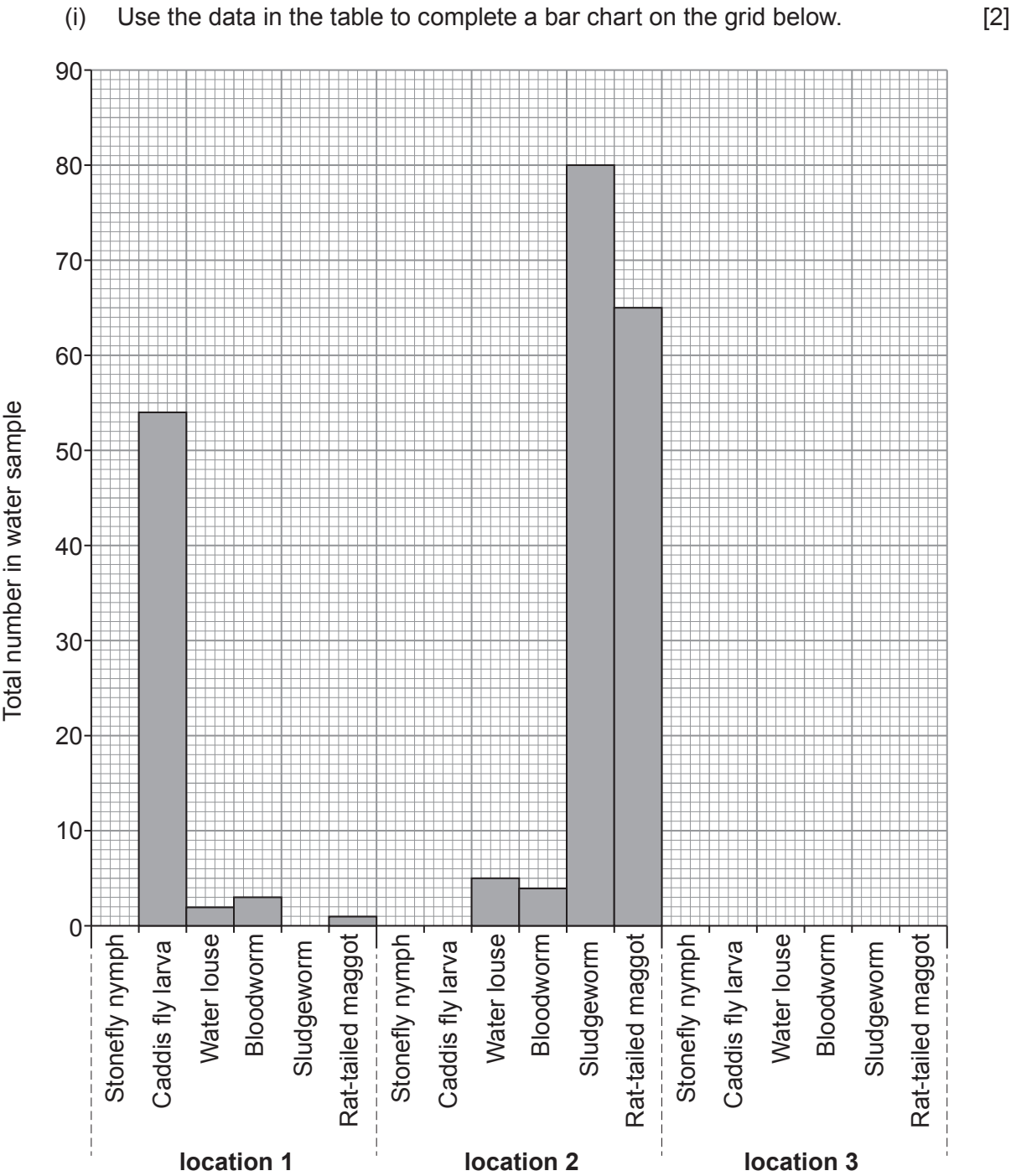


Results from the water sampling at each location are given in the table below.

Species	Total number in water sample		
	location 1	location 2	location 3
Stonefly nymph	0	0	31
Caddis fly larva	54	0	4
Water louse	2	5	3
Bloodworm	3	4	2
Sludgeworm	0	80	1
Rat-tailed maggot	1	65	1



Examiner
only



(ii) State which location is the cleanest. [1]

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(iii) Give **one** reason for your answer. [1]

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